

PD-RPM Viewed Through Three Systems Lenses

Three lenses, not three options. The three system types are not competing labels to choose among; they are nested levels of resolution. The peritoneal dialysis remote patient monitoring system (PD-RPM) is built out of complex systems, behaves as a system of systems, and must be managed as an enterprise system. Which lens applies depends on the question being asked, and using the wrong one is a reliable way to mismanage the program.

As a complex system. Kossiakoff defines a system as a set of interrelated components working together toward a common objective, with the distinguishing attributes of a single integrated system, a single acquisition authority, and a specific set of problems solved. The test is decomposition: take it apart and the pieces do nothing on their own. The home gateway qualifies. Its enclosure, radio, local storage, firmware, and power supply are useless individually, one organization procures and integrates them, and together they solve exactly one problem — collect readings and forward them reliably. The cyclor, the blood-pressure cuff, and the dashboard each qualify on the same terms. What does not qualify is PD-RPM as a whole: it has no single acquisition authority, which is exactly where this lens runs out.

As a system of systems. Maier defines a system of systems as an arrangement of systems related or connected to provide a given capability, where the loss of any part degrades the whole. PD-RPM satisfies all five attributes. Operational independence: the cyclor dialyzes the patient whether or not PD-RPM exists, and the EHR runs the clinic regardless. Managerial independence: the cyclor belongs to its manufacturer, the EHR to its vendor, the network to a carrier, each with separate funding and release cycles. Geographic distribution: patient homes, a regional monitoring center, and a data center. Emergent behavior: a firmware update that rennumbers alarm codes can silently break a threshold rule nobody thought to retest. Evolutionary development: there is no version one, only a baseline of whatever the constituents happen to be. The loss test holds as well — remove the vitals peripherals and monitoring continues, degraded.

Which kind of system of systems. Of Maier's four governance types, PD-RPM is an acknowledged system of systems. It has recognized objectives and a designated manager in the dialysis organization, but the constituent systems keep independent ownership, funding, and sustainment, so change happens by collaboration rather than direction. It is not directed, since no one can subordinate the cyclor manufacturer's roadmap, and not merely collaborative or virtual, since someone is accountable for the whole.

As an enterprise system. Rebovich and White define an enterprise as a purposeful combination of people who make decisions, their workflow process, and the technical systems supporting that decision-making. All five attributes are present: a common mission of keeping patients on home therapy; people, processes, and technology together, since the nurse's triage workflow and the nephrologist's prescription decision are inside the system rather than outside it; a network of multiple organizations spanning the dialysis provider, manufacturer, EHR vendor, carrier, supplier, and CMS; dispersed locations; and dependence on external resources. The decisive point is that PD-RPM's output is not data but a decision someone acts on.

Why the lens changes the engineering. Each lens names a different failure mode, and all three are live. Treating PD-RPM as a complex system implies authority the program does not have, and produces plans that assume the cyclor manufacturer will accommodate a schedule. The system-of-systems lens corrects that: leverage lies in agreements and interfaces, not directives, which is why interface management is a first-class work package rather than a design detail. The enterprise lens corrects both: a perfectly specified interface still fails if the nurse's workflow cannot absorb the alert volume it generates.

Where this lands for the project. For the remainder of the semester I will treat PD-RPM as an acknowledged system of systems, operating inside an enterprise, assembled from complex systems most of which this project does not own. That classification explains earlier decisions — why the work breakdown structure excludes the cyclor while defining an interface to it, and why the context diagram places it outside the boundary. It also constrains later architectures, which must be built around negotiated interfaces rather than assumed control.